3.3 ACETATE

Acetate fibres are chemical compounds of cellulose and not the pure cellulose as in rayon fibres. These chemical compounds have their own unique properties and in fact the various types of cellulose acetate fibres differ from each other in properties. In 1865, Pant Schutzenberger discovered the reaction of cellulose acetate. Later in 1903, the German chemists Arthur Eiechengeun and Theodore Becker invented the fast soluble form of cellulose acetate.

In 1904, cellulose acetate was developed by Camille Dreyfus and his younger brother Henri in England, during World War I. It was used as a non flammable lacquer for the fabrics used for covering the wings and fuselage of aircrafts. In 1918, they perfected the technique of spinning this substance into lustrous filaments of "artificial silk". Subsequently, British Celanese Ltd. started the production of this fibre. In 1924, the commercial production of acetate fabric started in the United States by Celanese Corp. In 1954, The Celanese Corp. of America undertook the production of a variation of this chemical called cellulose triacetate. This product is marketed by the Celanese Fibres Marketing Co. under the trademark Arnel.

3.3.1 Types of Acetate

The different types of acetate are

1. Diacetate
2. Triacetate.

3.3.2 Manufacturing Process of Acetate

Pure cellulose is formed by converting cotton linters or wood sheets. This pure cellulose is steeped in glacial acetic acid and aged under controlled temperature for a period of time. It is then mixed with acetic anhydride and a small amount of sulphuric acid as a catalyst to facilitate the reaction which produces a thick and clear liquid solution of cellulose acetate. This liquid is then passed through the spinneret to form these fibres which is solidified and rolled on the bobbins for later use (Figure 3.2).

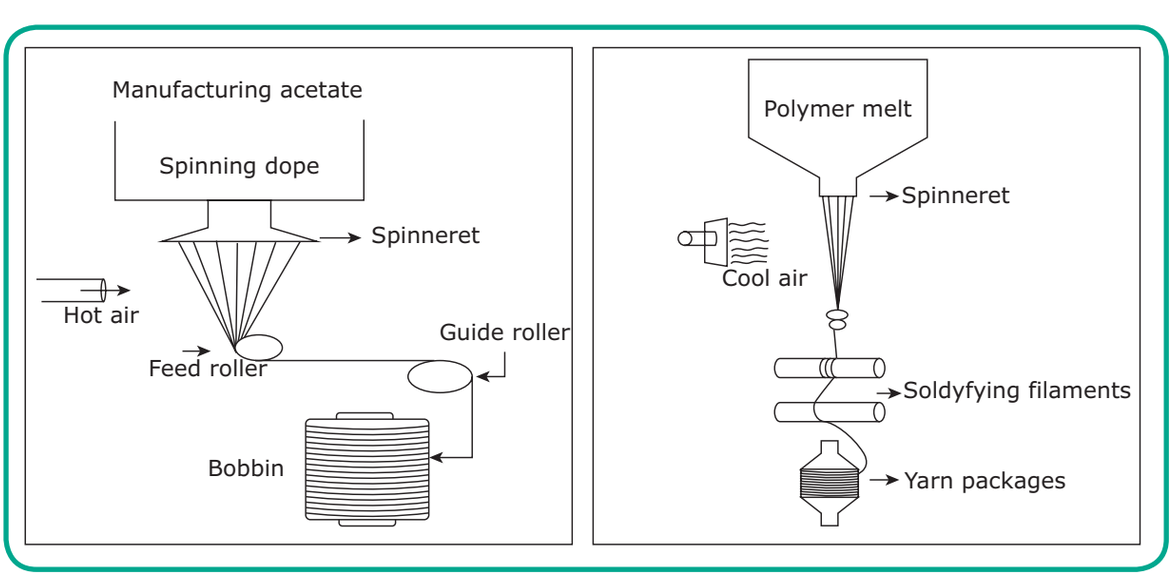


Figure 3.2 Manufacturing Process of Acetate

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